

EE6204 Systems Analysis — Practice Problems (Quiz 1)

Scope: Linear Programming — Graphic Method, Simplex Method, Sensitivity Analysis

Answer each question in the space provided. The answer key and full solutions are collected at the end of the document. All questions are taken from the course lecture notes.

Part A — Problem Formulation

Q1 Formulate the LP (Tailor's problem)

A tailor has the following materials available: 16 m² cotton, 11 m² silk, 15 m² wool. A suit requires 2 m² cotton, 1 m² silk, 1 m² wool. A gown requires 1 m² cotton, 2 m² silk, 3 m² wool. If a suit sells for \$30 and a gown for \$50, how many of each garment should the tailor make to obtain the maximum amount of money? **Formulate** the linear program.

Working space

Q2 Formulate the LP (Chemical products)

A person requires 10, 12 and 12 units of chemicals A, B and C respectively for his garden. A liquid product contains 5, 2, 1 units of A, B, C respectively per jar; a dry product contains 1, 2, 4 units respectively per carton. The liquid product sells for \$3 per jar and the dry product sells for \$2 per carton. How many of each should he purchase to minimize the cost and meet the requirements? **Formulate** the linear program.

Working space

Q3 Put the program into standard form

```
min    z = 5x1 + 3x2 - 7x3
s.t.    2x1 + 4x2 + 6x3 = 7
        3x1 - 5x2 + 3x3 ≤ 5
        -4x1 - 9x2 + 4x3 ≤ -4
        x1 ≥ -2, 0 ≤ x2 ≤ 4, x3 free
```

Convert to standard form: nonnegative right-hand sides, nonnegative variables, and equality constraints obtained by introducing slack / surplus variables as needed.

Working space

Q4 Standard matrix form (minimization)

```
min    z = 6x1 + 3x2 + 4x3
s.t.    x1 + 6x2 + x3 = 10
        2x1 + 3x2 + x3 = 15
        all variables nonnegative
```

Put the program into standard matrix form (define X , C , A , B , X_0) and state the initial basic feasible solution.

Working space

Q5 Standard matrix form (maximization)

$$\begin{array}{ll}\max & z = x_1 + x_2 \\ \text{s.t.} & x_1 + 5x_2 \leq 5 \\ & 2x_1 + x_2 \leq 4 \\ & x_1, x_2 \geq 0\end{array}$$

Put the program into standard matrix form (define X , C , A , B , X_0) and state the initial basic feasible solution.

Working space

Q6 Initial basic feasible solution by inspection

$$\begin{array}{ll}\min & z = 3x_1 + 5x_2 \\ \text{s.t.} & x_1 \leq 4 \\ & 2x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0\end{array}$$

Convert to standard form and write down the initial basic feasible solution.

Working space

Q7 Artificial variables and initial BFS

$$\begin{array}{ll}\min & z = 2x_1 + 3x_2 \\ \text{s.t.} & 3x_1 + 2x_2 = 14 \\ & 2x_1 - 4x_2 \geq 12 \\ & 4x_1 + 3x_2 \leq 19 \\ & x_1, x_2 \geq 0\end{array}$$

Introduce slack / surplus / artificial variables as necessary and give the initial basic feasible solution.

Working space

Q8 Solve graphically and enumerate BFS

$$\begin{array}{ll}\min & z = -2x_1 - x_2 \\ \text{s.t.} & 3x_1 + 4x_2 \leq 12 \\ & 2x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0\end{array}$$

- (a) Solve the problem graphically.
- (b) Enumerate all basic solutions and identify the basic feasible solutions (BFS).
- (c) State the optimal solution(s) and the optimal value.

Working space (graph + enumeration)

Part D — Simplex Method

Q9 Simplex method (minimization)

Use the simplex method to solve:

$$\begin{array}{ll}\min & z = -2x_1 - x_2 \\ \text{s.t.} & 3x_1 + 4x_2 \leq 12 \\ & 2x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0\end{array}$$

Working space (simplex tableaus)

Q10 Simplex method (maximization)

Use the simplex method to solve:

$$\begin{array}{ll}\max & z = x_1 + x_2 \\ \text{s.t.} & x_1 + 5x_2 \leq 5 \\ & 2x_1 + x_2 \leq 4 \\ & x_1, x_2 \geq 0\end{array}$$

Working space (simplex tableaus)

Q11 Two-phase method (with artificial variable)

Solve by the two-phase method:

$$\begin{array}{ll}\min & z = 80x_1 + 60x_2 \\ \text{s.t.} & 0.2x_1 + 0.32x_2 \leq 0.25 \\ & x_1 + x_2 = 1 \\ & x_1, x_2 \geq 0\end{array}$$

Working space (two-phase tableaus)

Q12 Infeasible program

Solve the following program, or show that it has no solution:

$$\begin{array}{ll}\min & z = 2x_1 + 3x_2 \\ \text{s.t.} & x_1 + 2x_2 \leq 2 \\ & 6x_1 + 4x_2 \geq 24 \\ & \text{all variables nonnegative}\end{array}$$

Working space

Part E — Duality

Q13 Write the symmetric dual (minimization primal)

Write the symmetric dual of the program:

```
min     $z = 5x_1 + 2x_2 + x_3$ 
s.t.    $2x_1 + 3x_2 + x_3 \geq 20$ 
        $6x_1 + 8x_2 + 5x_3 \geq 30$ 
        $7x_1 + x_2 + 3x_3 \geq 40$ 
        $x_1 + 2x_2 + 4x_3 \geq 50$ 
       all variables nonnegative
```

Working space

Q14 Write the symmetric dual (maximization primal)

Write the symmetric dual of the program:

```
max     $z = 2x_1 + x_2$ 
s.t.    $x_1 + 5x_2 \leq 10$ 
        $x_1 + 3x_2 \leq 6$ 
        $2x_1 + 2x_2 \leq 8$ 
       all variables nonnegative
```

Working space

Q15 Solve a program through its dual

Solve the following program by considering its dual:

```
min    z = 10x1 + 6x2 + 8x3
s.t.   x1 + x2 + 2x3 ≥ 2
       5x1 + 3x2 + 2x3 ≥ 1
       all variables nonnegative
```

Give both the dual and the primal optimal solutions.

Working space

Q16 Complementary slackness

Consider the primal

```
max    z = 60x1 + 30x2 + 20x3
s.t.   8x1 + 6x2 + x3 ≤ 48
       4x1 + 2x2 + 1.5x3 ≤ 20
       2x1 + 1.5x2 + 0.5x3 ≤ 8
       x1, x2, x3 ≥ 0
```

and its dual. The optimal primal solution is $(x_1^*, x_2^*, x_3^*, s_1^*, s_2^*, s_3^*) = (2, 0, 8, 24, 0, 0)$ and the optimal dual solution is $(w_1^*, w_2^*, w_3^*, e_1^*, e_2^*, e_3^*) = (0, 10, 10, 0, 5, 0)$, where s are primal slack variables and e are dual surplus variables. Verify the complementary slackness conditions.

Working space

Q17 Sensitivity: changes in the right-hand side B

$$\begin{aligned}
 \max \quad & z = 3x_1 + 5x_2 \\
 \text{s.t.} \quad & x_1 \leq 4 \\
 & 2x_2 \leq 12 \\
 & 3x_1 + 2x_2 \leq 18 \\
 & x_1, x_2 \geq 0
 \end{aligned}$$

The final simplex tableau gives the slack-coefficient matrix

$$\begin{aligned}
 S = & \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix}, \quad y^T = [0, 3/2, 1]
 \end{aligned}$$

and the optimal solution $x_1 = 2$, $x_2 = 6$, $z = 36$. Suppose $B = [4, 12, 18]^T$ is changed to $B_c = [4, 12+\Delta, 18]^T$. Find the range of Δ for which the solution remains feasible, and the corresponding range of b_2 .

Working space

Q18 Sensitivity: coefficient of a nonbasic variable

$$\begin{array}{ll}\max & z = 3x_1 + 5x_2 \\ \text{s.t.} & x_1 \leq 4 \\ & 2x_2 \leq 24 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0\end{array}$$

In the optimal solution, x_1 is a **nonbasic** variable. If the coefficient $c_1^* = 3$ is changed to $3 + \Delta$, find the allowable range to stay optimal (assume this is the only change in the model).

Working space

Q19 Sensitivity: coefficient of a basic variable

$$\begin{array}{ll}\max & z = 3x_1 + 3x_2 \\ \text{s.t.} & x_1 \leq 4 \\ & 3x_2 \leq 24 \\ & 3x_1 + 4x_2 \leq 18 \\ & x_1, x_2 \geq 0\end{array}$$

In the optimal solution, x_2 is a **basic** variable. If the coefficient $c_2^* = 3$ is changed to $3 + \Delta$, find the allowable range to stay optimal (assume this is the only change in the model).

Working space

Answer Key & Solutions

Full solutions for all questions. Work through the problems before checking.

Q1 Tailor's problem (formulation)

Let x = number of suits and y = number of gowns.

Maximize $f = 30x + 50y$
subject to $2x + y \leq 16$ (cotton)
 $x + 2y \leq 11$ (silk)
 $x + 3y \leq 15$ (wool)
 $x, y \geq 0$ and integer

Answer: the LP above. (If solved, the integer optimum is $x = 7, y = 2$ with $f = \$310$.)

Q2 Chemical products (formulation)

Let x = number of jars (liquid) and y = number of cartons (dry).

Minimize $F = 3x + 2y$
subject to $5x + y \geq 10$ (chemical A)
 $2x + 2y \geq 12$ (chemical B)
 $x + 4y \geq 12$ (chemical C)
 $x, y \geq 0$

Answer: the LP above.

Q3 Standard form

Step 1. Multiply the third constraint by -1 so every right-hand side is nonnegative:

$$4x_1 + 9x_2 - 4x_3 \geq 4$$

Step 2. The free variable becomes the difference of two nonnegative variables: $x_3 = x_3' - x_3''$.

Step 3. The bound $x_1 \geq -2$ is handled by substituting $x_1 = x_1' - 2$, i.e. $x_1' = x_1 + 2$ (≥ 0). The upper bound $x_2 \leq 4$ is kept as a general constraint.

$$\begin{aligned} \min \quad & z' = 5x_1' + 3x_2 - 7x_3' + 7x_3'' \quad (z' = z + 10) \\ \text{s.t.} \quad & 2x_1' + 4x_2 + 6x_3' - 6x_3'' = 11 \\ & 3x_1' - 5x_2 + 3x_3' - 3x_3'' + x_4 = 11 \quad (x_4 = \text{slack}) \\ & 4x_1' + 9x_2 - 4x_3' + 4x_3'' - x_5 = 4 \quad (x_5 = \text{surplus}) \\ & x_2 + x_6 = 4 \quad (x_6 = \text{slack}) \\ & x_1', x_2, x_3', x_3'', x_4, x_5, x_6 \geq 0 \end{aligned}$$

Answer: the standard form above.

Q4 Standard matrix form (minimization)

Add an artificial variable to each equality constraint that has no slack variable, with penalty M :

$$\begin{aligned} \min \quad & z = 6x_1 + 3x_2 + 4x_3 + \overline{Mx_4} + \overline{Mx_5} \\ \text{s.t.} \quad & x_1 + 6x_2 + x_3 + \overline{x_4} = 10 \\ & 2x_1 + 3x_2 + x_3 + \overline{x_5} = 15 \\ & \text{all variables nonnegative} \end{aligned}$$

$$\begin{aligned} X &= [x_1 \ x_2 \ x_3 \ \overline{x_4} \ \overline{x_5}]^T \\ C &= [6 \ 3 \ 4 \ M \ M]^T \\ A &= \begin{bmatrix} 1 & 6 & 1 & 1 & 0 \\ 2 & 3 & 1 & 0 & 1 \end{bmatrix} \\ B &= [\overline{10} \ \overline{15}]^T \\ X_0 &= [\overline{x_4} \ \overline{x_5}]^T \end{aligned}$$

Initial BFS: $\overline{x_4} = 10$, $\overline{x_5} = 15$, $x_1 = x_2 = x_3 = 0$.

Q5 Standard matrix form (maximization)

$$\begin{array}{ll}\max & z = x_1 + x_2 + 0x_3 + 0x_4 \\ \text{s.t.} & x_1 + 5x_2 + x_3 = 5 \\ & 2x_1 + x_2 + x_4 = 4 \\ & \text{all variables nonnegative}\end{array}$$
$$\begin{aligned}X &= [x_1 \ x_2 \ x_3 \ x_4]^T \\ C &= [1 \ 1 \ 0 \ 0]^T \\ A &= \begin{bmatrix} 1 & 5 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix} \\ B &= [5 \ 4]^T \\ X_0 &= [x_3 \ x_4]^T\end{aligned}$$

Initial BFS: $x_3 = 5, x_4 = 4, x_1 = x_2 = 0$.

Q6 Initial BFS by inspection

$$\begin{array}{ll}\min & z = 3x_1 + 5x_2 \\ \text{s.t.} & x_1 + x_3 = 4 \\ & 2x_2 + x_4 = 12 \\ & 3x_1 + 2x_2 + x_5 = 18 \\ & \text{all variables nonnegative}\end{array}$$

There are 5 variables and 3 constraints (2 degrees of freedom). Set the nonbasic variables $x_1 = x_2 = 0$ and read off the basic (slack) variables.

Initial BFS: $(x_1, x_2, x_3, x_4, x_5) = (0, 0, 4, 12, 18)$.

Q7 Artificial variables

First introduce surplus / slack variables (e = surplus, s = slack):

$$\begin{array}{lll}3x_1 + 2x_2 & & = 14 \\ 2x_1 - 4x_2 - e & & = 12 \\ 4x_1 + 3x_2 & + s & = 19\end{array}$$

Equations (1) and (2) have no slack variable, so add artificial variables to them:

$$\begin{array}{llll}3x_1 + 2x_2 & + a_3 & = 14 \\ 2x_1 - 4x_2 - e + a_5 & & = 12 \\ 4x_1 + 3x_2 & + s_6 & = 19 \\ \text{all variables nonnegative}\end{array}$$

Initial BFS: $(x_1, x_2, e_4, a_3, a_5, s_6) = (0, 0, 0, 14, 12, 19)$.

Q8 Graphic method & BFS

Standard form: $\min z = -2x_1 - x_2 + 0x_3 + 0x_4$, with $3x_1 + 4x_2 + x_3 = 12$, $2x_1 + x_2 + x_4 = 6$. Since $n - m = 4 - 2 = 2$, each basic solution sets 2 variables to zero. There are ${}^4C_2 = 6$ basic solutions:

$x_1=x_2=0 \rightarrow (0, 0, 12, 6)$	BFS ($x_3=12, x_4=6$)
$x_2=x_4=0 \rightarrow (3, 0, 3, 0)$	BFS
$x_1=x_4=0 \rightarrow (0, 3, 0, 3)$	BFS
$x_3=x_4=0 \rightarrow (12/5, 6/5, 0, 0)$	BFS
$x_1=x_3=0 \rightarrow (4, 0, 0, -2)$	infeasible ($x_4 = -2$)
$x_2=x_3=0 \rightarrow (0, 6, -12, 0)$	infeasible ($x_3 = -12$)

The 4 BFS correspond to the 4 extreme points $(0,0)$, $(3,0)$, $(0,3)$, $(12/5, 6/5)$.

$(0, 0)$	: $z = 0$
$(3, 0)$	: $z = -2(3) - 0 = -6$
$(0, 3)$	: $z = -2(0) - 3 = -3$
$(12/5, 6/5)$	: $z = -2(12/5) - 6/5 = -30/5 = -6$

Minimum $z = -6$, attained at $(3, 0)$ and $(12/5, 6/5)$ and at every point on the segment joining them (infinitely many optimal solutions).

Q9 Simplex (minimization)

Standard form: $\min z = -2x_1 - x_2 + 0x_3 + 0x_4$. The initial (min) tableau has last row $C^T - C_0^T A = [-2, -1, 0, 0]$:

	x_1	x_2	x_3	x_4	RHS	
x_3	3	4	1	0	12	
x_4	2	1	0	1	6	
	-2	-1	0	0	0	most negative: -2 (x_1 column)
ratio:	$12/3 = 4$,	$6/2 = 3$	$\rightarrow$ pivot = 2			

	x_1	x_2	x_3	x_4	RHS	
x_3	0	$5/2$	1	$-3/2$	3	
x_1	1	$1/2$	0	$1/2$	3	
	0	0	0	1	6	no negative entries in last row

$x_1^* = 3, x_3^* = 3, x_2^* = x_4^* = 0$, with $z^* = -6$ (minimization: z^* is the negative of the last-row/last-column entry).

Q10 Simplex (maximization)

Standard form: $\max z = x_1 + x_2 + 0x_3 + 0x_4$. Initial (max) tableau has last row $C_0^T A - C^T = [-1, -1, 0, 0]$:

	x1	x2	x3	x4	RHS
x3	1	5	1	0	5
x4	2	1	0	1	4
	-1	-1	0	0	0
ratio:	5/1 = 5, 4/2 = 2 → pivot = 2				

After two pivots the final tableau is:

	x1	x2	x3	x4	RHS
x2	0	1	2/9	-1/9	2/3
x1	1	0	-1/9	5/9	5/3
	0	0	1/9	4/9	7/3

$x_1^* = 5/3, x_2^* = 2/3, x_3^* = x_4^* = 0$, with $z^* = 7/3$ (maximization).

Q11 Two-phase method

Standard matrix form with $X = [x_1 \ x_2 \ x_3 \ \overline{x_4}]^T$, $C = [80 \ 60 \ 0 \ M]^T$, $A = [[0.2 \ 0.32 \ 1 \ 0], [1 \ 1 \ 0 \ 1]]$, $B = [0.25 \ 1]^T$, $X_0 = [x_3 \ \overline{x_4}]^T$, $C_0 = [0 \ M]^T$.

Last row $C^T - C_0^T A = [80-M, 60-M, 0, 0]$, $C_0^T B = M$, which is split into two rows for Phase 1 / Phase 2. Driving out the artificial variable and then pivoting gives:

$$x_1^* = 0.583, \quad x_2^* = 0.4166, \quad x_3^* = \overline{x_4}^* = 0$$

$z^* = 71.667$ (minimization).

Q12 Infeasible program

Introduce a slack variable x_3 in the first constraint and a surplus variable x_4 plus an artificial variable $\overline{x_5}$ in the second:

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 2 \\ 6x_1 + 4x_2 - x_4 + \overline{x_5} &= 24 \end{aligned}$$

Running the two-phase method, the next-to-last row reaches optimality (no negative entry above a zero of the last row), but the artificial variable $\overline{x_5}$ remains basic with nonzero value.

By rule C5, the original program has no solution (infeasible).

Q13 Symmetric dual (min primal)

The primal is $\min z = C^T X, AX \geq B, X \geq 0$. Its dual is $\max z = B^T W, A^T W \leq C, W \geq 0$:

$$\begin{array}{ll}\max & z = 20w_1 + 30w_2 + 40w_3 + 50w_4 \\ \text{s.t.} & 2w_1 + 6w_2 + 7w_3 + w_4 \leq 5 \\ & 3w_1 + 8w_2 + w_3 + 2w_4 \leq 2 \\ & w_1 + 5w_2 + 3w_3 + 4w_4 \leq 1 \\ & w_1, w_2, w_3, w_4 \geq 0\end{array}$$

Answer: the dual above.

Q14 Symmetric dual (max primal)

The primal is $\max z = C^T X, AX \leq B, X \geq 0$. Its dual is $\min z = B^T W, A^T W \geq C, W \geq 0$:

$$\begin{array}{ll}\min & z = 10w_1 + 6w_2 + 8w_3 \\ \text{s.t.} & w_1 + w_2 + 2w_3 \geq 2 \\ & 5w_1 + 3w_2 + 2w_3 \geq 1 \\ & w_1, w_2, w_3 \geq 0\end{array}$$

Answer: the dual above.

Q15 Solve through the dual

The primal (min, 2 constraints) has a dual with only 2 variables, which is easier to solve:

$$\begin{array}{ll}\max & z = 2w_1 + w_2 \\ \text{s.t.} & w_1 + 5w_2 \leq 10 \\ & w_1 + 3w_2 \leq 6 \\ & 2w_1 + 2w_2 \leq 8 \\ & w_1, w_2 \geq 0\end{array}$$

Solving the dual by the simplex method gives:

$$w_1^* = 4, \quad w_2^* = 0, \quad z^* = 8$$

The primal solution is read from the last row of the dual's final tableau, in the columns of the slack variables:

$$x_1^* = 0, \quad x_2^* = 0, \quad x_3^* = 1, \quad z^* = 8$$

Dual optimum $(w_1, w_2) = (4, 0)$; primal optimum $(x_1, x_2, x_3) = (0, 0, 1)$; $z^* = 8$.

Q16 Complementary slackness

For a max primal with $\leq$ constraints, the complementary slackness conditions are $s_i^* w_i^* = 0$ (primal slack $\times$ dual variable) and $e_j^* x_j^* = 0$ (dual surplus $\times$ primal variable).

$$\begin{aligned} s1^* w1^* &= 24 \cdot 0 = 0 \\ s2^* w2^* &= 0 \cdot 10 = 0 \\ s3^* w3^* &= 0 \cdot 10 = 0 \\ e1^* x1^* &= 0 \cdot 2 = 0 \\ e2^* x2^* &= 5 \cdot 0 = 0 \\ e3^* x3^* &= 0 \cdot 8 = 0 \end{aligned}$$

All six products equal zero, so the complementary slackness conditions hold.

Q17 Changes in B

Feasibility of the (same) basis is maintained iff $S \cdot Bc \geq 0$:

$$\begin{aligned} S \cdot Bc &= \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix} \begin{bmatrix} 4 \\ 12+\Delta \\ 18 \end{bmatrix} \\ &= \begin{bmatrix} (6+\Delta)/3 \\ (12+\Delta)/2 \\ (6-\Delta)/3 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} (6+\Delta)/3 \geq 0 &\Rightarrow \Delta \geq -6 \\ (12+\Delta)/2 \geq 0 &\Rightarrow \Delta \geq -12 \\ (6-\Delta)/3 \geq 0 &\Rightarrow \Delta \leq 6 \end{aligned}$$

$-6 \leq \Delta \leq 6$, i.e. $6 \leq b2 \leq 18$.

Q18 Nonbasic coefficient

Here $y^T = [0, 0, 5/2]$ and the column of the nonbasic variable $x1$ is $A1^* = [1, 0, 3]^T$. Optimality is preserved iff the corresponding reduced cost stays nonnegative:

$$\begin{aligned} y^T A1^* - (3 + \Delta) &= [0 \ 0 \ 5/2] \cdot [1 \ 0 \ 3]^T - (3 + \Delta) \\ &= 15/2 - 3 - \Delta \\ &= 9/2 - \Delta \geq 0 \end{aligned}$$

$\Delta \leq 9/2$, i.e. $c1 \leq 15/2$.

Q19 Basic coefficient

Here $y^T = [3/4, 0, 3/4]$ and the column of the basic variable x_2 is $A_2^* = [0, 3, 4]^T$. First recompute the reduced cost, then restore the proper Gaussian form (the x_2 column must be $[0, 0, 1]^T$).

$$\begin{aligned} y^T A_2^* - (3 + \Delta) &= [3/4 \ 0 \ 3/4] \cdot [0 \ 3 \ 4]^T - (3 + \Delta) \\ &= 3 - 3 - \Delta = -\Delta \end{aligned}$$

After restoring Gaussian form, the coefficients of the nonbasic variables x_3 and x_5 in the last row are:

$$\begin{aligned} \text{for } x_3: \quad -3/4 \cdot \Delta + 3/4 &\geq 0 \Rightarrow \Delta \leq 1 \\ \text{for } x_5: \quad 1/4 \cdot \Delta + 3/4 &\geq 0 \Rightarrow \Delta \geq -3 \end{aligned}$$

$$\mathbf{-3 \leq \Delta \leq 1, \text{ i.e. } 0 \leq c_2 \leq 4.}$$